

ST117 Individual DRAFT Written Report - Phase 2 Task 1

(Woodland Vegetation - Tree Measurements)

Part a

```
vw_wide <- VW %>%
  filter(FIELDNAME %in% c("DIAMETER", "HEIGHT", "DISTANCE", "NUM_STEMS")) %>%
  pivot_wider(
    id_cols = c(SITECODE, SYEAR, PLOTPID, CELLID, TREEID, STEMID, TREE_SPEC),
    names_from = FIELDNAME,
    values_from = VALUE,
    values_fn = first
  ) %>%
  mutate(across(c(DIAMETER, HEIGHT, DISTANCE, NUM_STEMS), as.numeric))

cols <- intersect(names(vw_wide), c("SITECODE", "SYEAR", "TREE_SPEC", "DIAMETER", "HEIGHT"))
vw_wide %>%
  select(all_of(cols)) %>%
  head(3) %>%
  knitr::kable()
```

SITECODE	SYEAR	TREE_SPEC	DIAMETER	HEIGHT
T05	1994	237	19	25
T05	1994	237	16	22
T05	1994	237	23	25

Part b

```
# Use all sites, remove NA
lm_all <- lm(HEIGHT ~ DIAMETER, data = vw_wide)
# Create a table of coefficients
lm_all %>% tidy() %>% knitr::kable(caption = "Coefficients for Height ~ Diameter (all sites)")
```

Table 2: Coefficients for Height ~ Diameter (all sites)

term	estimate	std.error	statistic	p.value
(Intercept)	9.1393251	0.2282432	40.04205	0
DIAMETER	0.1036816	0.0065583	15.80930	0

Based on the positive DBH-height relationship, we fitted a simple linear regression (Height ~ Diameter). The slope (0.104) means that for a 1 cm increase in DBH, height increases by 0.104 m on average. The model is statistically significant ($p < 2e-16$) but explains only 11% of the variance in height ($R^2=0.11$), indicating a weak linear relationship.

Part c

```
## c) Site-separated models with diagnostics and comparison

vw_clean <- vw_wide %>% filter(!is.na(DIAMETER) & !is.na(HEIGHT))

sites <- vw_clean %>%
```

```

group_by(SITECODE) %>%
summarise(n = n()) %>%
filter(n >= 10) %>%
pull(SITECODE)

selected_sites <- sites[1:min(4, length(sites))]

models <- list()
scatter_plots <- list()
resid_plots <- list()

lm_global <- lm(HEIGHT ~ DIAMETER, data = vw_clean)
models[["Global"]] <- lm_global

p_global_scatter <- ggplot(vw_clean, aes(x = DIAMETER, y = HEIGHT)) +
  geom_point(alpha = 0.2, size = 0.8) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(title = "Global model (all sites)", x = "DBH (cm)", y = "Height (m)") +
  theme_minimal()
scatter_plots[["Global"]] <- p_global_scatter

resid_global <- data.frame(fitted = fitted(lm_global), resid = residuals(lm_global))
p_global_resid <- ggplot(resid_global, aes(x = fitted, y = resid)) +
  geom_point(alpha = 0.2, size = 0.8) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(title = "Global: Residuals vs Fitted", x = "Fitted values", y = "Residuals") +
  theme_minimal()
resid_plots[["Global"]] <- p_global_resid

for (site in selected_sites) {
  data_site <- vw_clean %>% filter(SITECODE == site)
  lm_site <- lm(HEIGHT ~ DIAMETER, data = data_site)
  models[[site]] <- lm_site

  p_scatter <- ggplot(data_site, aes(x = DIAMETER, y = HEIGHT)) +
    geom_point(alpha = 0.5) +
    geom_smooth(method = "lm", se = FALSE, color = "red") +
    labs(title = paste("Site", site), x = "DBH (cm)", y = "Height (m)") +
    theme_minimal()
  scatter_plots[[site]] <- p_scatter

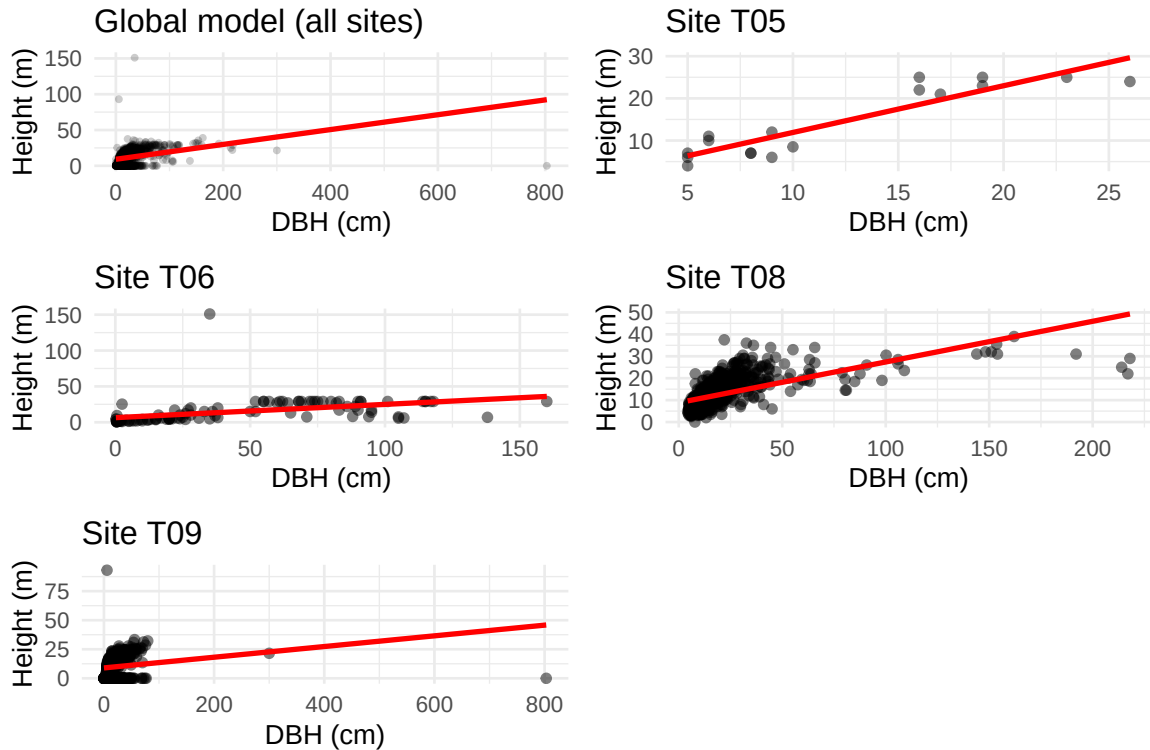
  resid_df <- data.frame(fitted = fitted(lm_site), resid = residuals(lm_site))
  p_resid <- ggplot(resid_df, aes(x = fitted, y = resid)) +
    geom_point(alpha = 0.5) +
    geom_hline(yintercept = 0, linetype = "dashed") +
    labs(title = paste("Site", site, "- Residuals"), x = "Fitted", y = "Residuals") +
    theme_minimal()
  resid_plots[[site]] <- p_resid
}

all_scatter <- c(list(Global = p_global_scatter), scatter_plots[selected_sites])
all_resid <- c(list(Global = p_global_resid), resid_plots[selected_sites])

grid.arrange(grobs = all_scatter, ncol = 2, nrow = ceiling(length(all_scatter)/2),
  top = "Height vs Diameter: Global and site-specific models")

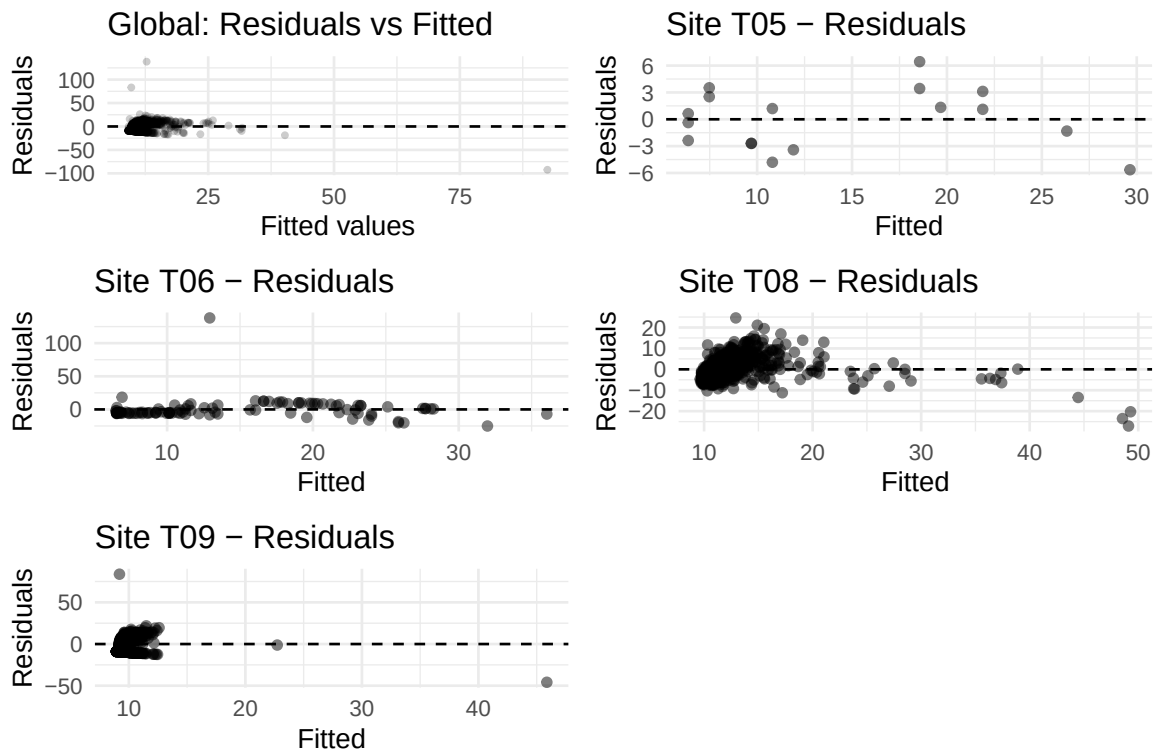
```

Height vs Diameter: Global and site-specific models



```
grid.arrange(grobs = all_resid, ncol = 2, nrow = ceiling(length(all_resid)/2),
             top = "Residual vs Fitted plots")
```

Residual vs Fitted plots



```
comparison_table <- data.frame(
  Model = names(models),
  R_squared = sapply(models, function(m) summary(m)$r.squared),
  Adj_R_sq = sapply(models, function(m) summary(m)$adj.r.squared),
  RMSE = sapply(models, function(m) sigma(m)),
  AIC = sapply(models, function(m) AIC(m)),
  BIC = sapply(models, function(m) BIC(m)),
  n = sapply(models, function(m) nobs(m))
)
```

```
)
kable(comparison_table, caption = "Model fit comparison", digits = 3)
```

Table 3: Model fit comparison

	Model	R_squared	Adj_R_sq	RMSE	AIC	BIC	n
Global	Global	0.114	0.113	8.049	13678.263	13694.991	1951
T05	T05	0.839	0.829	3.417	93.897	96.396	17
T06	T06	0.173	0.165	15.983	858.837	866.712	102
T08	T08	0.395	0.394	5.062	5252.111	5266.392	863
T09	T09	0.026	0.025	8.351	6867.024	6881.653	969

The global model (all sites pooled) explains only 11.4% of the variance in height ($R^2=0.114$) and has a high RMSE (8.05 m). This weak fit suggests that a single linear relationship between DBH and height does not hold across different woodland sites, likely due to site-specific factors (species composition, age structure, management).

Among the four site-specific models, T05 stands out with an excellent fit ($R^2=0.839$, RMSE=3.42 m) despite having only 17 trees. The high R^2 indicates a strong linear association between DBH and height. However, the small sample size makes the estimate fragile, and the relatively narrow DBH range may inflate the apparent model fit.

T06 has a weak fit ($R^2=0.173$, RMSE=16.0 m). The high RMSE and low R^2 suggest substantial unexplained variation, while the residual plot indicates heteroscedasticity and possible non-linearity. This may reflect variation in species composition, age structure, or environmental conditions within the site.

T08 shows a moderate fit ($R^2=0.3955$, RMSE=5.06 m). The relationship is significant but still leaves >60% of variation unexplained. The moderate RMSE suggests that linear regression can capture a general trend, but predictions will have substantial error.

T09 performs worst ($R^2=0.026$, RMSE=8.35 m), indicating almost no linear relationship between DBH and height. The scatterplot suggests substantial variability in height across similar DBH values, implying that additional ecological or site-specific factors may influence tree height.

AIC/BIC generally favour the better-fitting site-specific models, particularly T05. However, these criteria should be interpreted cautiously because the models are fitted to datasets with very different sample sizes, making direct comparisons across sites less meaningful.

Conclusion: The global model masks substantial variation between woodland sites, so site-specific models provide a more informative description of the DBH–height relationship. T05 shows a strong linear trend but is limited by its small sample size, whereas T06 and T09 display weak fits and evidence of heteroscedasticity or non-linearity. Overall, the results highlight considerable spatial heterogeneity in tree allometry across ECN woodland sites.

Part d

```
## d) Identify complications and propose improvements
```

```
### 1. Identify one issue per site
```

```
complication_table <- data.frame(
  Site = c("T05", "T06", "T08", "T09"),
  Issue = c(
    "Very small sample size (n=17) - high leverage and overfitting risk",
    "Heteroscedasticity: variance of height increases with DBH (fan shape in residuals)",
    "Non-linearity: height plateaus for large DBH (residual curvature)",
    "Weak linear relationship ( $R^2$  near 0); height varies substantially across similar DBH values"
  ),
  Proposed_remedy = c(
    "Use only descriptive statistics; refrain from inference",
    "Weighted least squares (weights =  $1/DBH^2$ )",
```

```

    "Polynomial regression: add quadratic term",
    "No linear regression is sensible; use summary statistics or log transformation (still poor)"
  )
)
knitr::kable(complication_table, caption = "Complications and suggested remedies per site")

```

Table 4: Complications and suggested remedies per site

Site	Issue	Proposed_remedy
T05	Very small sample size (n=17) - high leverage and overfitting risk	Use only descriptive statistics; refrain from inference
T06	Heteroscedasticity: variance of height increases with DBH (fan shape in residuals)	Weighted least squares (weights = 1/DBH ²)
T08	Non-linearity: height plateaus for large DBH (residual curvature)	Polynomial regression: add quadratic term
T09	Weak linear relationship (R ² near 0); height varies substantially across similar DBH values	No linear regression is sensible; use summary statistics or log transformation (still poor)

2. Improve T08 with quadratic term

```

data_T08 <- vw_clean %>% filter(SITECODE == "T08")
lm_T08_lin <- lm(HEIGHT ~ DIAMETER, data = data_T08)
lm_T08_quad <- lm(HEIGHT ~ DIAMETER + I(DIAMETER^2), data = data_T08)

# Compare fits
comparison_T08 <- data.frame(
  Model = c("Linear", "Quadratic"),
  R_squared = c(summary(lm_T08_lin)$r.squared, summary(lm_T08_quad)$r.squared),
  AIC = c(AIC(lm_T08_lin), AIC(lm_T08_quad)),
  RMSE = c(sigma(lm_T08_lin), sigma(lm_T08_quad))
)
knitr::kable(comparison_T08, caption = "T08: Linear vs Quadratic fit", digits = 4)

```

Table 5: T08: Linear vs Quadratic fit

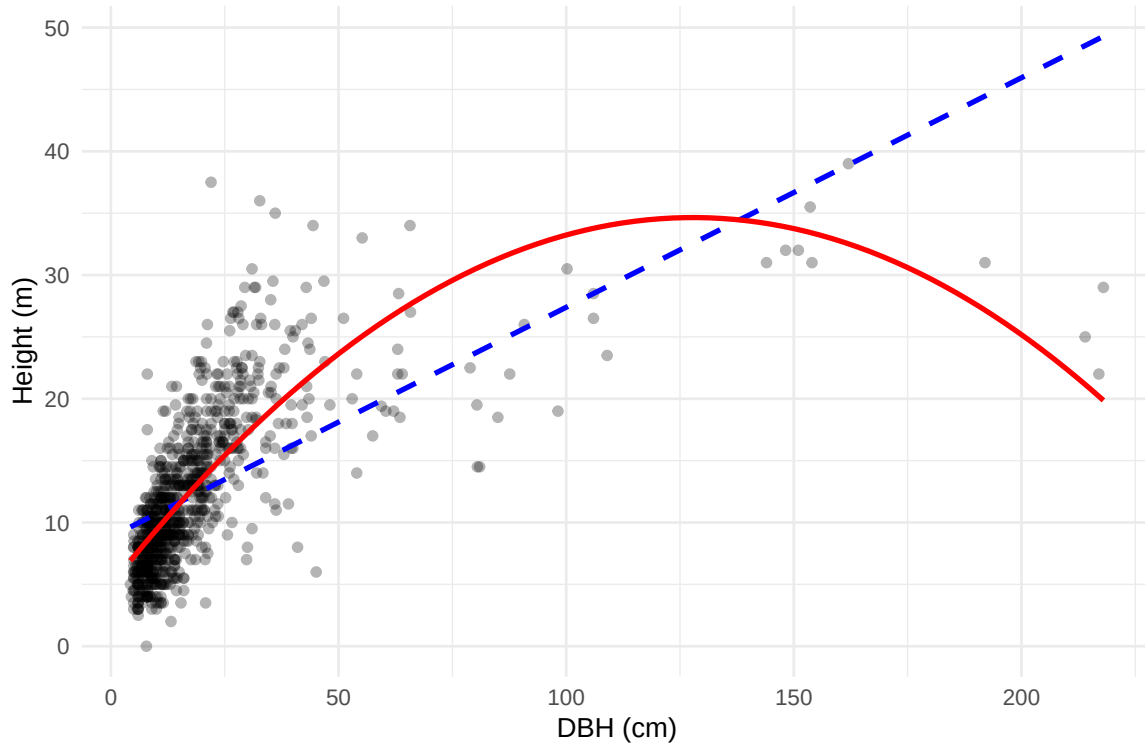
Model	R_squared	AIC	RMSE
Linear	0.3951	5252.111	5.0616
Quadratic	0.5770	4945.438	4.2352

```

# Plot both fits
p_T08 <- ggplot(data_T08, aes(x = DIAMETER, y = HEIGHT)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE, color = "blue", linetype = "dashed") +
  geom_smooth(method = "lm", formula = y ~ x + I(x^2), se = FALSE, color = "red") +
  labs(title = "T08: Linear (blue dashed) vs Quadratic (red)", x = "DBH (cm)", y = "Height (m)") +
  theme_minimal()
print(p_T08)

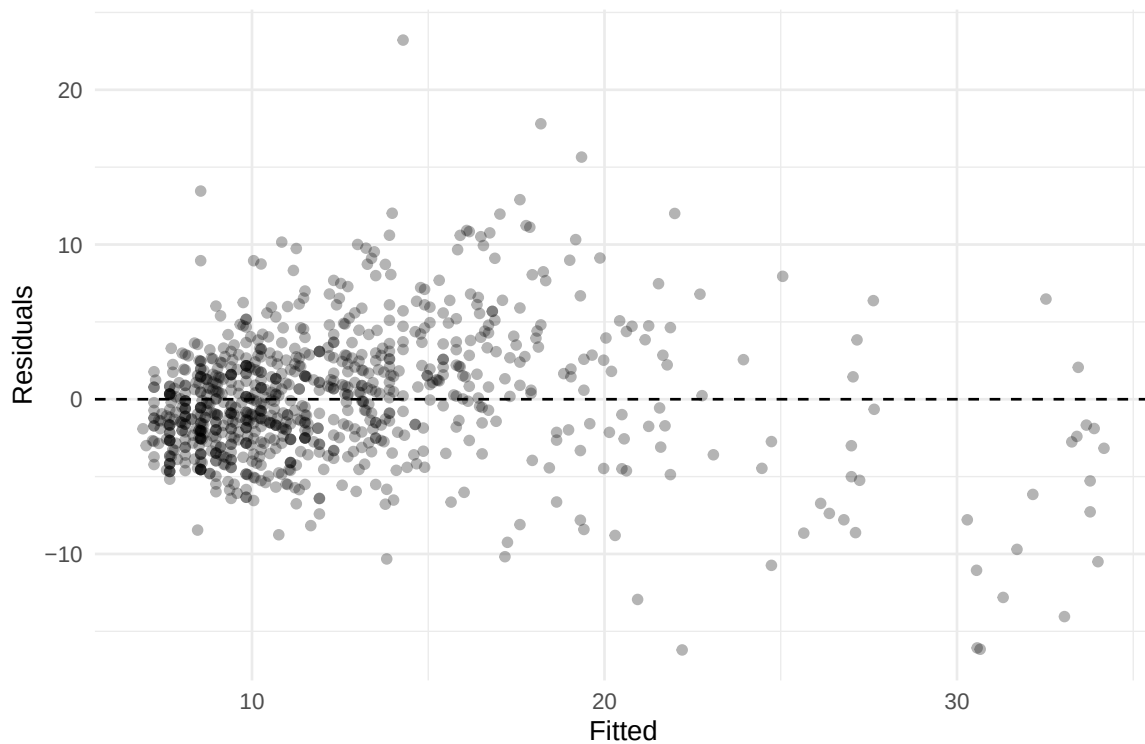
```

T08: Linear (blue dashed) vs Quadratic (red)



```
# Residual plot for quadratic model
resid_T08 <- data.frame(fitted = fitted(lm_T08_quad), resid = residuals(lm_T08_quad))
p_resid_T08 <- ggplot(resid_T08, aes(x = fitted, y = resid)) +
  geom_point(alpha = 0.3) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(title = "T08 Quadratic model: Residuals vs Fitted", x = "Fitted", y = "Residuals") +
  theme_minimal()
print(p_resid_T08)
```

T08 Quadratic model: Residuals vs Fitted



```
### 3. Improve T06 with weighted least squares (WLS)
```

```

data_T06 <- vw_clean %>% filter(SITECODE == "T06")
# Ordinary LS
lm_T06_ols <- lm(HEIGHT ~ DIAMETER, data = data_T06)
# Weighted LS: weight = 1/DIAMETER^2 (to downweight large DBH which have higher variance)
weights <- 1 / (data_T06$DIAMETER^2)
lm_T06_wls <- lm(HEIGHT ~ DIAMETER, data = data_T06, weights = weights)

# Compare
comparison_T06 <- data.frame(
  Model = c("OLS", "WLS (1/DBH^2)"),
  R_squared = c(summary(lm_T06_ols)$r.squared, summary(lm_T06_wls)$r.squared),
  AIC = c(AIC(lm_T06_ols), AIC(lm_T06_wls)),
  RMSE = c(sigma(lm_T06_ols), sigma(lm_T06_wls))
)
knitr::kable(comparison_T06, caption = "T06: OLS vs Weighted Least Squares", digits = 4)

```

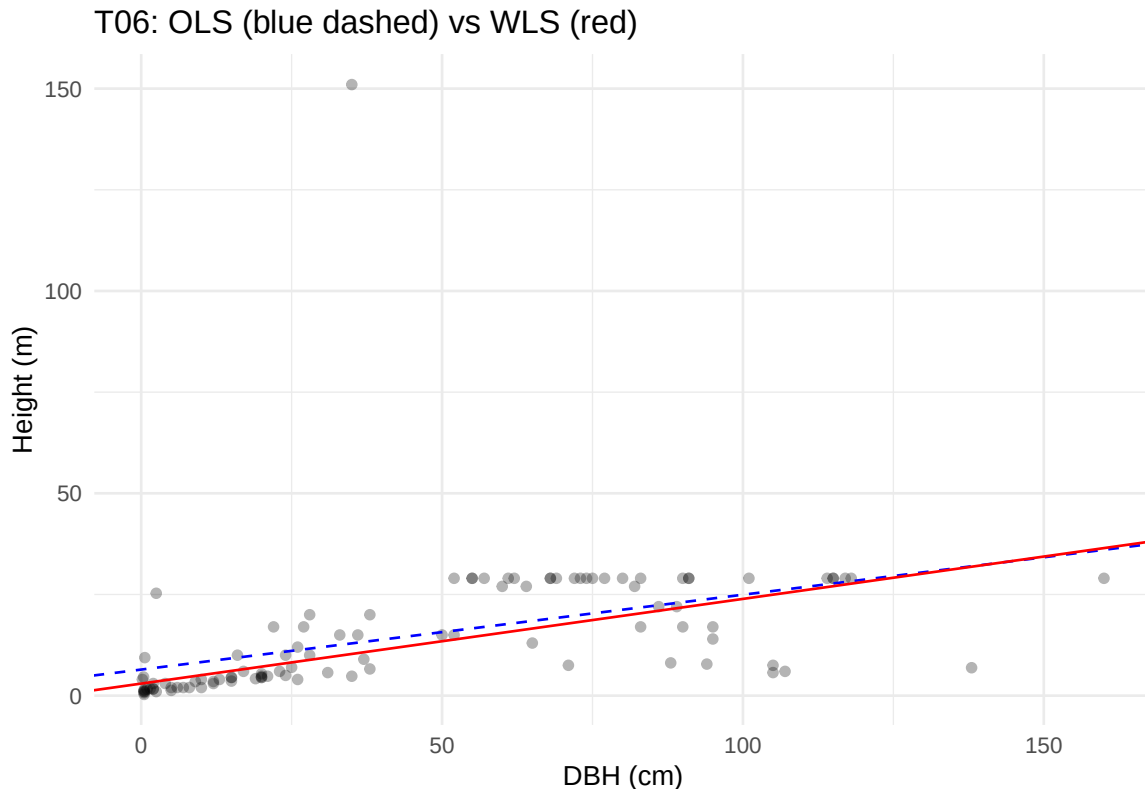
Table 6: T06: OLS vs Weighted Least Squares

Model	R_squared	AIC	RMSE
OLS	0.1729	858.8372	15.9832
WLS (1/DBH^2)	0.0106	1049.3761	1.9008

```

# Scatter with two regression lines
p_T06 <- ggplot(data_T06, aes(x = DIAMETER, y = HEIGHT)) +
  geom_point(alpha = 0.3) +
  geom_abline(intercept = coef(lm_T06_ols)[1], slope = coef(lm_T06_ols)[2], color = "blue", linetype = "dashed") +
  geom_abline(intercept = coef(lm_T06_wls)[1], slope = coef(lm_T06_wls)[2], color = "red") +
  labs(title = "T06: OLS (blue dashed) vs WLS (red)", x = "DBH (cm)", y = "Height (m)") +
  theme_minimal()
print(p_T06)

```

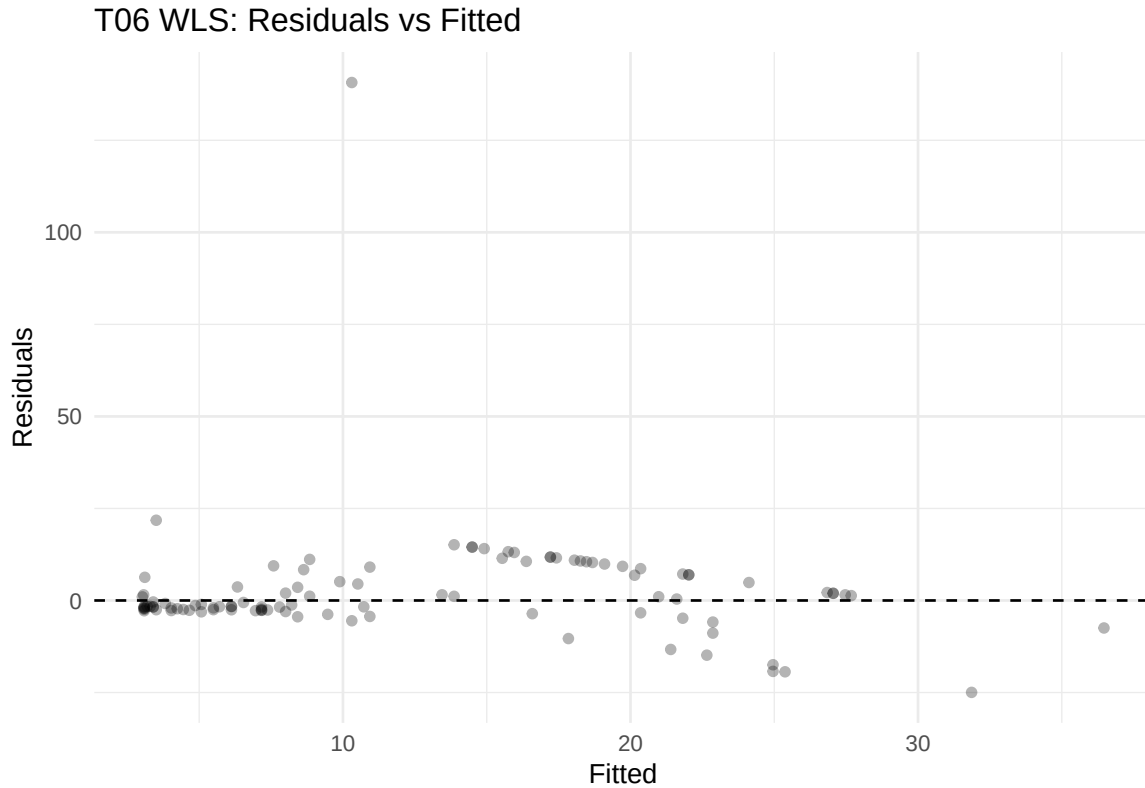


```

# Residual plot for WLS
resid_T06_wls <- data.frame(fitted = fitted(lm_T06_wls), resid = residuals(lm_T06_wls))
p_resid_T06 <- ggplot(resid_T06_wls, aes(x = fitted, y = resid)) +
  geom_point(alpha = 0.3) +

```

```
geom_hline(yintercept = 0, linetype = "dashed") +
labs(title = "T06 WLS: Residuals vs Fitted", x = "Fitted", y = "Residuals") +
theme_minimal()
print(p_resid_T06)
```



Improved fit for T08: Adding the quadratic term increased R^2 from 0.395 to 0.577, reduced AIC from 5252 to 4945, and reduced RMSE from 5.06 to 4.24. The residual plot shows less curvature than the linear model, indicating that the quadratic model better captures the non-linear relationship between DBH and height for larger trees.

Improved fit for T06: Weighted least squares (weights = $1/\text{DBH}^2$) reduced the influence of large DBH observations with higher variance. Although the weighted model produced a much lower residual standard error, the R^2 value remained low and the AIC increased relative to the OLS model. Nevertheless, the residual plot shows a more constant spread across fitted values, suggesting improved handling of heteroscedasticity.

Sites T05 and T09: For T05, the small sample size makes any regression unstable; we only report descriptive statistics. For T09, no linear or simple quadratic model fits well; a non-linear model (e.g., log-log) would be required but is beyond the scope of ST117. We therefore conclude that simple linear regression is not sensible for T09.